

Fertility, Immigration, and the Housing Supply*

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Abstract

We develop a political economy model of the housing market in which the supply of new homes is shaped by a democratic vote among households. Voter support for additional housing is determined by the effect this vote has on households' expected discounted utility via changes in house prices. Households that own housing, and therefore typically experience financial losses from additional supply, generally oppose new housing: the 'Not In My Backyard' (NIMBY) effect. Households that wish to buy a house and would benefit from cheaper housing support more supply. We find that support for additional supply is highest among households that are younger, low-income or are currently renting. We then simulate the impact of a temporary baby boom in the population. We find that those born during the baby boom attain higher rates of homeownership and greater net worth due to their greater political influence relative to other generations. However, this comes at the expense of higher house prices, lower homeownership rates and less wealth for the subsequent generations. Calibrating the model to US demographic data, we find this political economy mechanism was responsible for a 15 percent increase in house prices since 1950. Using demographic projections we forecast that due to population aging house prices will continue to rise into the 21st century.

[FERTILITY DIFFERENCE]This draft keeps the v13 paper structure almost line-for-line and only marks project-03 changes. The added objective is to test whether an initial baby-boom shock, together with immigration composition, predicts a subsequent fertility fall through housing-cost feedback.

Abstract

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1 The Model

In this section we outline a life-cycle model with housing and debt choices. The model features heterogeneous, life-cycle households that optimize their consumption of non-durable consumption, housing and saving in liquid assets. A number of factors influence households preferences on house prices and thus their political support for additional housing supply. These include household income and wealth as well as portfolio allocation and age over the life cycle. We use the model to study the housing market response to an exogenous change in demographics. Following the shock, the behavior of owner-occupiers and renters determines equilibrium house prices.

A consumer of age j enters the economy at age 25 and lives for further J periods and maximizes lifetime expected utility which consists of consumption c_t , housing services, h_t and a bequest $v(w_{J+1})$. Housing services depends on whether they own or rent the property. Individuals also have access to a liquid asset that pays an interest r_{t+1} . The household's lifetime utility function is given by

$$\sum_{j=0}^J \beta^j u(c_j, h_j) + \beta^{J+1} v(w_{J+1}) \quad (1)$$

where households optimize over consumption, c_t , liquid assets, a_t , and housing, h_t at each age j . The flow utility is increasing and strictly concave in c and h . Households receive utility from wealth they bequest w_{J+1} . We assume the functional form

$$u(c_t, h_t) = \frac{(c_t^{1-\zeta} h_t^\zeta)^{1-\gamma}}{1-\gamma} \quad (2)$$

where the consumption of housing stock consists of either the amount of housing owned or rented

Relative to Gross and Chivers (2025), we augment the household block with an endogenous fertility margin. The fertility-extension flow utility used in the model note is

$$u_t = \frac{(c_t^{1-\zeta} h_{t,eff}^\zeta)^{1-\gamma}}{1-\gamma} + \chi \frac{n_{t,home}^{1-\eta}}{1-\eta} \quad (3)$$

with effective housing services

$$h_{t,eff} = \frac{h_t}{(1 + \lambda n_{t,home})^\psi}. \quad (4)$$

Hence, additional children reduce effective space through crowding, and this feeds back into housing demand and fertility timing.

Children-at-home is a state variable generated by cohort dynamics:

$$N_{0,t+1} = b_t, \quad N_{a+1,t+1} = N_{a,t}, \quad n_{t,home} = \sum_{a=0}^{A_{leave}-1} N_{a,t}. \quad (5)$$

where b_t are births and A_{leave} is the leave-home age.

In the current prototype code, fertility choice is implemented with the reduced-form shortcut

$$f_t^* = \text{clip} \left(f_{\text{level}} \exp \left(-\eta_p p_t^h \right), 0.05, 0.85 \right), \quad b_t = f_t^* M_t^{15-44} \mu_t, \quad (6)$$

which is intended as a computational approximation to the full crowding-based utility mechanism above.

where M_t^{15-44} is the female population age 15-44 and μ_t is the temporary baby-boom multiplier.

$$h_t = \begin{cases} \theta^{rent} h_t^{rent} & \text{if } h_t^{own} = 0 \\ h_t^{own} & \text{if } h_t^{own} > 0. \end{cases} \quad (7)$$

The curvature parameter γ determines households' risk aversion and inter-temporal elasticity of substitution and ζ denotes housing's share of total consumption. Rental housing provides a stream of utility that is discounted by θ^{rent} compared to self-owned housing. Finally there is a warm glow bequest function when households die that provides utility

$$v(w) = \psi \frac{(w + \bar{w})^{1-\gamma}}{1-\gamma} \quad (8)$$

which generates a bequest function with decreasing marginal utility with respect to wealth, w , governed by the parameters $\bar{w}$ and ψ . Households supply one unit of labor inelastically and are paid a wage $y_{t,j}$. The wage they receive varies according to their productivity which is driven by the exogenous process

$$\ln y_{t,j} = g_j + z_t \text{ for } j \leq J^{retired} \quad (9)$$

where z_t is a Markov process calibrated to yield realistic inequality (as per Kindermann and Krueger (2022)) and g_j is a deterministic age profile. The stochastic process comprises of an AR(1) process for lowest income states, with two additional top-income states which incorporate "superstar" incomes. Once workers reach retirement age $J^{retired}$ income falls to a fixed level based on their wage upon retirement. The income process is normalized to have a mean of 1, such that the price of housing p_t^h reflects the house price to income ratio.

Households may save or borrow liquid assets subject to a borrowing constraint where each

household can borrow against a proportion, m , of the total value of the stock of housing they own

$$a_t \geq -mh_t^{own}p_t^h. \quad (10)$$

Households may buy or sell housing at price, p_t^h , subject to a minimum block size $\bar{h}$. All property sales, including those that are sold as part of a bequest are subject to a transaction cost F_{sell} proportional to the total value of property sold, which is reflected by the indicator function I_{sell} . If they do not own any housing they must rent it from real estate firms at price p_t^{rent} . The household's utility function is thus maximized subject to the general budget constraint¹

$$a_{t+1} = (1 + r_t^a) a_t + y_t - c_t - p_t^{rent} h_t^{rent} - \left(1 - I^{sell} F^{sell}\right) p_t^h \Delta h_t^{own} \quad (11)$$

where I^{sell} is an indicator variable for households that choose to sell part or all of their current stock of housing.

We assume the households start with some initial wealth with the mean of the distribution of initial liquid wealth, a_{25} , defined by the average bequest of the household's at the end of their life.

1.1 Household's Problem

Each household's state vector is $\mathbf{s}_t = \{a_t, h_{t-1}^{own}, z_t\}$, where a_t is the stock of liquid assets, h_{t-1}^{own} is the amount of housing they owned at the end of period $t-1$ and z_t is the persistent component of their income. The price of housing, p_t , is an aggregate state variable. The value function for the household is given by the maximum of the three possible housing tenure options: rent (either because they owned no housing or sell it all that period), or are owners who choose to adjust their stock of housing or maintain their current level of housing. Households who choose to rent (and thus sell whatever stock of housing they owned at the beginning of the period) are able to freely adjust the amount of housing they rent each period. The value function for a household that choose to rent housing (and thus sell any previously owned housing stock) at age j is

$$\begin{aligned} V_j^{rent}(\mathbf{s}_t; p_t) &= \max_{c_t, a_{t+1}, h_t^{rent}} u(c_t, h_t^{rent}) + \beta E_t V_{j+1}(\mathbf{s}_{t+1}; p_{t+1}) \\ &\quad s.t. \\ a_{t+1} &= (1 + r_t^a) a_t + y_t - c_t - p_t^{rent} h_t^{rent} + \left(1 - F^{sell}\right) p_t^h h_{t-1}^{own} \\ a_{t+1} &\geq 0 \\ h_t^{own} &= 0 \end{aligned} \quad (12)$$

¹Noting that households only interact with the rental market if they own no housing stock of their own.

The value function for home owners who choose not to adjust their (non-zero) stock of housing in period t is

$$\begin{aligned}
V_j^{non-adjust}(\mathbf{s}_t; p_t) &= \max_{c_t, a_{t+1}} u(c_t, h_t^{own}) + \beta E_t V_{j+1}(\mathbf{s}_{t+1}; p_{t+1}) \\
&\quad s.t. \\
a_{t+1} &= (1 + r_t^a) a_t + y_t - c_t \\
a_{t+1} &\geq -m p_t^h h_t^{own} \\
h_t^{own} &= h_{t-1}^{own}
\end{aligned} \tag{13}$$

finally, the value function for households who choose to buy or sell housing in period t is

$$\begin{aligned}
V_j^{adjust}(\mathbf{s}_t; p_t) &= \max_{c_t, a_{t+1}, h_t^{own}} u(c_t, h_t^{own}) + \beta E V_{j+1}(\mathbf{s}_{t+1}; p_{t+1}) \\
&\quad s.t. \\
a_{t+1} &= (1 + r_t^a) a_t + y_t - c_t - \left(1 - I^{sell} F^{sell}\right) p_t^h \Delta h_t^{own} \\
a_{t+1} &\geq -m p_t^h h_t^{own} \\
h_t^{own} &= h_{t-1}^{own} + \Delta h_t^{own}
\end{aligned} \tag{14}$$

The final value function is the maximum of each of the three choices over housing tenure

$$V_j(\mathbf{s}_t; p_t) = \max \left\{ V_j^{rent}(\mathbf{s}_t; p_t), V_j^{adjust}(\mathbf{s}_t; p_t), V_j^{non-adjust}(\mathbf{s}_t; p_t) \right\} \tag{15}$$

The expected value function for the final year of life is the same for all types of housing tenure and is given by the warm glow bequest function given by equation 4 where total wealth is the sum of liquid and illiquid wealth given the current housing price.

$$w = a + \left(1 - F^{sell}\right) p_t^h h^{own} \tag{16}$$

1.2 Real Estate Firms

Real estate firms borrow from households by issuing debt in the form of the liquid asset to fund the purchase of housing, which is rented out to households that do not own their own home. Real estate firms operate in a perfectly competitive market, and thus the rental price is pinned down by the cost of borrowing, a fixed cost of providing rental services, the price of purchasing house stock and any expected capital appreciation driven by the drift in the price of housing. We assume that real estate firms are able to borrow against the full value of their housing stock. The zero profit condition pins down the price of rental stock as the function of the cost of housing,

capital gains from housing and the interest rate

$$p_t^{rent} = \phi^{rent} + (r_t^a) p_t^h - \Delta p_{t+1}^h, \quad (17)$$

where ϕ^{rent} is the per unit cost of providing rental services.

1.3 Housing Supply

The aggregate economy consists of unit measure of neighborhoods across which all types of households are equally distributed. The neighborhood's housing supply in period $t+1$ is determined in each neighborhood by a simple majoritarian voting mechanism in which households vote over the amount of permitted housing in an area. Only households who currently reside in that neighborhood can vote to determine the next periods level of new housing. Once the quantity of housing permits has been determined by the households' collective vote within the neighbourhood, the supply of housing is costlessly produced by a perfectly competitive construction sector in $t+1$. This is similar to Kaplan et al. (2020), who model a perfectly competitive construction sector in which all rents from land permits accrue to the government.

An increase in the housing supply will reduce the price of housing in the locality and households vote understanding that an increase in housing permits will lead to a small persistent decrease in the cost of housing. An increase in the housing supply will also affect the price of rental properties by lowering the input price for the real estate firms. As a result, each household in a neighborhood will vote to increase (or decrease) the availability of housing permits in their locality based on how a small, but permanent, change in house prices will affect their expected welfare. Specifically, households will vote to decrease the supply of housing if a small change in house prices, ϵ^p , increases their expected lifetime utility such that

$$V_j(\mathbf{s}; p^h + \epsilon^p) - V_j(\mathbf{s}; p^h) + \sigma \geq 0 \quad (18)$$

where σ represents a non-financial preference over the supply of housing. A negative or positive σ indicates a non-financial preference for lower or higher housing supply, respectively. For example, households may view new housing negatively as a result of congestion effects on infrastructure, or positively because of the new amenities that greater density could bring. Households may also hold ideological views about housing supply, which this parameter would capture as well.

In principle these preferences could vary by household, either due to changes in age, productivity, or as a result of an idiosyncratic shock. However, allowing for such heterogeneity would require introducing a larger set of novel parameters that would then need to be calibrated. To limit our degrees of freedom, we therefore calibrate these preferences as a simple, fixed constant across all households regardless of age, income or wealth. As a result, this parameter is effectively the average non-financial preference for all households. We test this assumption in Section 6 and find that the main results of the model do not substantively change with alternative assumptions.

The impact of a small perturbation to the price of housing will vary depending on household's ownership of liquid assets, housing and their current level of productivity. If a small increase in the price of housing increases their expected lifetime utility, then the left hand side of equation 14 will be positive and thus they will vote for lower supply of housing permits.

1.4 The Median Voter Theorem

Applying the median voter theorem, equilibrium in the housing market will occur when the median voter is indifferent to a change in the price of housing.² The equilibrium house price p_t is determined by the median voter equation

$$\int g(\mathbf{s}; p_t^h) \Theta(\mathbf{s}; p^h) ds = 0.5 \quad (19)$$

and

$$\Theta(\mathbf{s}; p^h) = \mathbf{1}_{\{V_j(\mathbf{s}; p^h + \epsilon^p) - V_j(\mathbf{s}; p^h) + \sigma \geq 0\}} \quad (20)$$

in which $g(\mathbf{s}; p_t^h)$ is the distribution of households over their state variables, and $\Theta(\mathbf{s}; p^h)$ is an indicator variable which defines whether a vote prefers higher house prices is equal to 0.5.

The change in expected lifetime utility from a small, persistent change in the price of housing is determined by the households current portfolio of assets and expected life-time income. Figure 1 shows how different household characteristics correlate with the decision to vote for lower supply and thus higher house prices. Households who currently rent or expect to have a high likelihood of purchasing new housing in the near term will vote for housing supply to increase thus decreasing the price of housing and relaxing their budget constraint. These households tend to be younger, have low incomes and low wealth. Looking at the distribution of voters according to the expected probability that a given household will make a purchase of housing at some point in their life, we can see that households that do not have a high expected probability of purchasing housing over the rest of their life-time will generally prefer a higher price of housing in order to increase the size of their bequest and the resale value of their stock of housing. By contrast the coalition of support for lower house prices consists of two distinct groups of households: households that anticipate buying additional housing (particularly in the near term) and thus want to loosen their budget constraint. as well as households that are currently renters because an increase in the supply of housing also lowers the price of rental units.

The equilibrium price of housing will thus vary according to the relative size of these groups of households. An exogenous increase in young households, who disproportionately rent and thus prefer lower house prices, will lead to an increase in the supply of housing. The price of

²While these preferences are in general well defined, it is important to note that preferences over house prices are not monotonic. For example, a household with a moderate amount of housing might prefer a small increase in house prices as this will increase their expected bequest, but would be better off if house prices fall substantially as they would be able to consume more housing.

housing will thus fall until a sufficient number of renters become homeowners, and accordingly switch their political support towards higher house prices, thus making the median household indifferent to a change in house prices.

In order to abstract from strategic voting across dynasties, we assume different children and parents live in different neighborhoods and thus do not effect the other's house market dynamics.³ Consequently even if an individual may benefit from a larger inheritance if the price of their parents home increases, it will not affect their vote on the optimal level of housing supply in their local neighborhood.⁴

We assume that households are unable to forecast future changes in political support for additional housing supply and instead assume that the house prices follow a random walk. This ensures the dynamics of the model are solvable as it does not require the households to consider the possibility of strategic voting (i.e. voting to lower the house price today to ensure wider support for higher house prices in future periods). We also assume that households are unable to form coalitions when voting, nor bargain political support across time periods and vote solely based on their individual interest at time t . Once the quantity of housing permits has been determined by the households' collective vote the supply of housing is costlessly supplied by a perfectly competitive construction sector. This is similar to Kaplan et al (2020) who assume a perfectly competitive construction sector in which all rents from land permits accrue to the government.⁵

³According to Fu (2019) 61.5% adults live more than 10 miles away from their mother.

⁴This assumption also makes our model robust to other forms of altruistic giving. For example, in addition to receiving a warm glow from giving to their descendants at death, households may care about their descendant's welfare directly and thus may wish to sell housing in order to give their children funds to purchase housing earlier. As families live in different neighbourhoods, however, this would still not change their voting behaviour. This is because parents with housing would still wish to restrict housing supply in their neighbourhood in order to receive the highest return on their house so they can maximise their child's welfare.

⁵This assumption allows us to abstract from any potential employment benefits derived from additional demand for the construction sector.

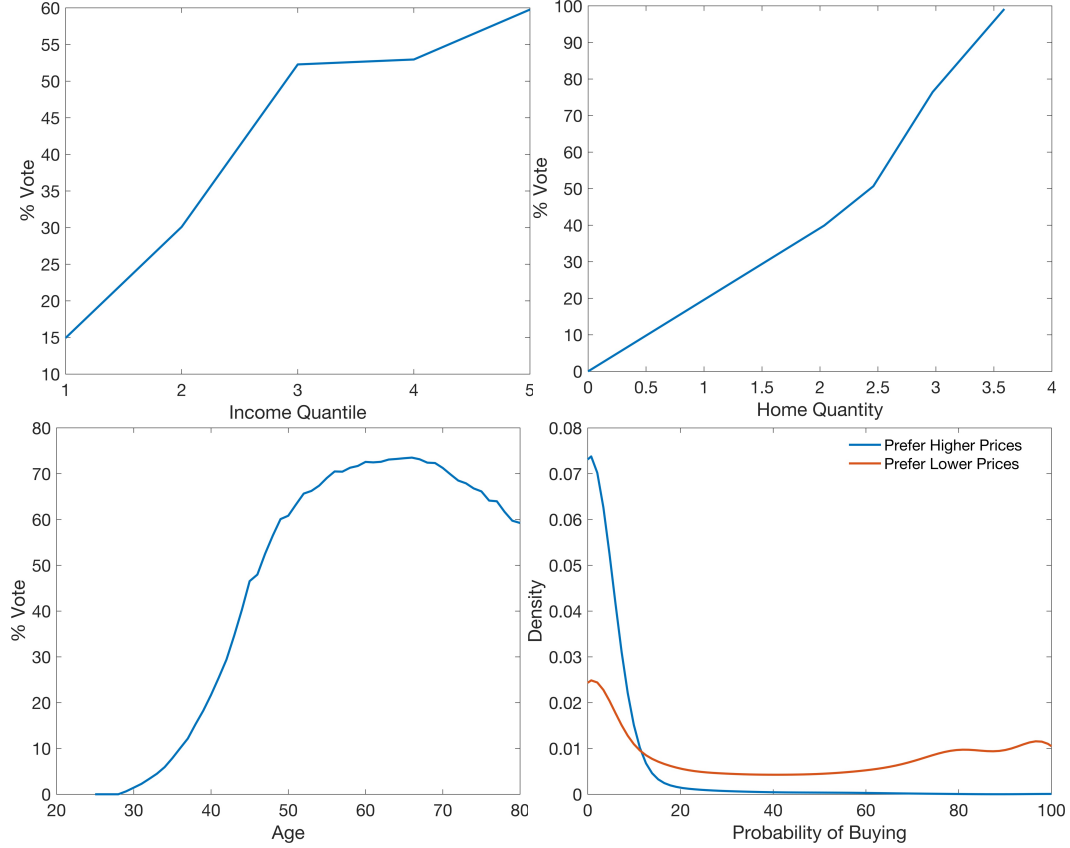


Figure 1: Household Support for Higher House Prices

1.5 Equilibrium

For ease of notation we use the distribution of households age $\Omega = \{\omega_{25}, \dots, \omega_{80}\}$, where ω_i is the share of the population at age i .⁶ The political-economic equilibrium of this model consists of value functions $\{V_j^{rent}(\mathbf{s}; p^h(\Omega)), V_j^{adjust}(\mathbf{s}; p^h(\Omega)), V_j^{non-adjust}(\mathbf{s}; p^h(\Omega))\}$, decision rules over the consumption, $c(\mathbf{s}; p^h(\Omega))$, housing demand, $\{h^{rent}(\mathbf{s}; p^h(\Omega)), h(\mathbf{s}; p^h(\Omega))\}$, liquid savings $a(\mathbf{s}; p^h(\Omega))$, and votes over housing supply, $\Theta(\mathbf{s}; p^h(\Omega))$, which define the households' optimal allocations as a function of their individual state variables and the aggregate price of housing, and the price of housing and rental units which in turn is a function of the distribution of the households' age $p^h(\Omega)$ and $p^{rent}(\Omega)$. After calibrating the age distribution of households across the model we set the formation of new households to keep the demographic structure constant (in the absence of demographic shocks). Finally we assume households expectations of the future path of house prices follow a random walk and thus in expectation will remain at their current level - at least with respect to the demographic changes in the model. The distribution of households, $g(\mathbf{s}; p^h(\Omega))$, is thus defined by the deterministic age and income processes, the

⁶We drop the t subscript for simplicity.

stochastic income shocks and their optimal choices in response to these shocks. The full numerical computational method is outlined in detail in the Appendix.

1.6 Project-03 differences (highlighted)

Beyond the utility extension, project 03 changes two additional model blocks relative to the baseline NIMBY write-up. First, political tightness includes a lagged baby-boom cohort term. Second, children-at-home feeds directly into housing demand dynamics.

$$\theta_t = \text{clip}(\theta_0 + \theta_{old} \cdot old_t - \theta_{young} \cdot young_t + \theta_{boom} \cdot \bar{b}_{t-\ell:t-\ell-w+1}, 0, 0.95) \quad (21)$$

$$D_t = d_0 + d_{young} \cdot young_t + d_{birth} \cdot b_t + 0.15 \cdot n_{t,home} \quad (22)$$

Coding alignment status (current MATLAB prototype): (i) the reduced-form fertility rule and lagged-boom political term above are coded; (ii) the full crowding-based household optimization problem is not yet solved as a full dynamic-programming household block. So the code is directionally aligned with the mechanism, but not yet a one-to-one implementation of the full structural utility equations.

1.7 Prototype MATLAB simulation evidence

Temporary baby-boom experiment used in the current code: +60% births for periods $t = 15$ to $t = 24$, damping 0.25, post window $t \geq 40$. Phase-delta results: new model $\Delta f_{post} = -0.001812$ and $\Delta p_{post} = +0.038142$; old proxy $\Delta f_{post} = -0.000107$ and $\Delta p_{post} = -0.085626$.

